CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

FACULTY OF TECHNOLOGY AND ENGINEERING

SUBJECT: WEB DEVELOPMENT FRAMEWORKS (ITUE203)

SEMESTER: 3RD, 2026-27 (ODD)

Practical - 11

Student Management CRUD with Search and Filter

Problem Definition: Create a student management module to add, view, edit, delete, search and filter student records. All database operations must use MySQLi statements and clear success/failure messages.

A. Theory

1. What is CRUD?

Answer: CRUD stands for Create, Read, Update, and Delete. These operations cover the basic lifecycle of database records.

2. Why use MySQLi prepared statements?

Answer: Prepared statements help keep user-provided values separate from SQL syntax and reduce SQL injection risk.

3. How do search and filter differ?

Answer: Search matches a user-entered phrase across selected fields. A filter restricts results using a condition such as course or status.

4. Why confirm deletion?

Answer: Confirmation helps prevent accidental removal. The server must still validate the record ID and permissions.

B. Implementation / Procedure

1. Database setup

Create a students table with id, name, email, course, phone, and created_at. Add suitable constraints, including a unique email if appropriate.

2. Create and Read

Use validated POST input and prepared INSERT statements to add records. Use SELECT to display students in a table.

3. Update

Load the selected record by ID, populate an edit form, validate submitted changes, and execute a prepared UPDATE statement.

4. Delete

Confirm the action in the interface, then send a POST request. Validate the ID and execute a prepared DELETE statement only for authorized users.

5. Search and filter

Use prepared SELECT queries with LIKE conditions for name/email/course and an optional course filter. Escape displayed values with htmlspecialchars().

C. Test Cases

Test Case	Input / Action	Expected Result
Add student	Submit valid student details	New record appears in the list
Edit student	Change course or phone	Updated values appear
Delete student	Confirm delete for a record	Record is removed and success message appears
Search	Search a known name/email	Only matching records are shown
Filter	Choose a course	Only students in that course are shown

D. Key Questions / Analysis / Interpretation

1. Are SELECT, INSERT, UPDATE, and DELETE operations working?

Answer: SELECT displays records, INSERT adds a student, UPDATE edits an existing record, and DELETE removes a record after confirmation. Each operation reports success or failure.

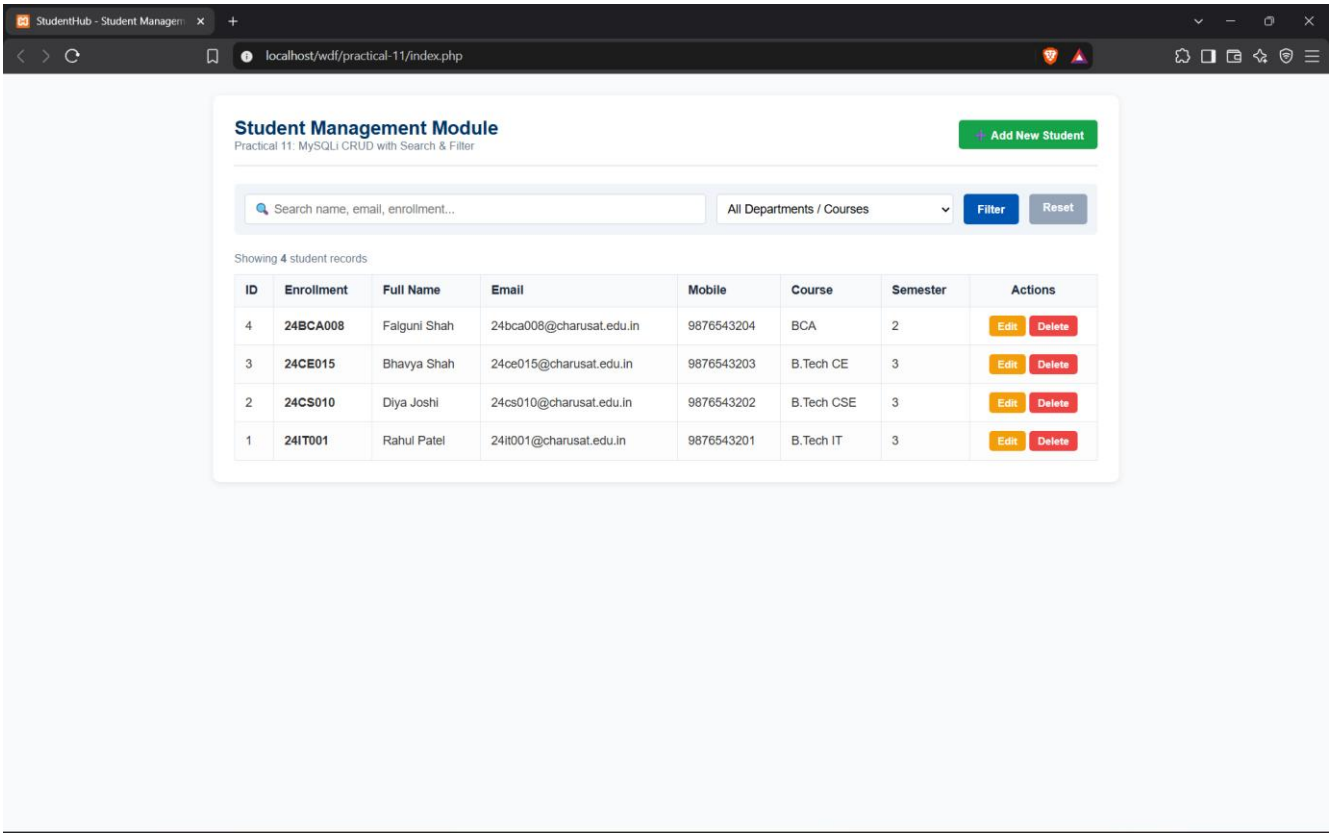
2. Is search by name/email/course implemented?

Answer: A search field can match name, email, or course using a prepared query. A course filter can narrow the results further.

3. Are delete operations confirmed before execution?

Answer: Yes. A confirmation prompt is shown before the delete request is submitted to prevent accidental deletion.

Work (Screenshots)



GitHub Repository Link

https://github.com/dhyandesai1727/25DIT012-WDF-BATCH-A